

Rishi Preetham Maddela

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence and Full-Stack Development, with hands-on experience building scalable, data-driven web applications using JavaScript, Python, Next.js, and PostgreSQL. Proficient in REST API development, database schema design, and AI-powered system integration. Strong foundation in data structures, algorithms, and object-oriented programming, with a focus on delivering production-quality software solutions.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C, PHP, Haskell

Web Technologies: Next.js, React, HTML5, CSS3, REST APIs

Databases: PostgreSQL, SQL, Database Schema Design

Developer Tools: Git, GitHub, VS Code, Eclipse IDE, MATLAB

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming (OOP), Functional Programming

PROJECTS

Repository Risk Intelligence Automation | *Node.js, GitHub API, AI Agents, Slack API, Email Automation* 2026

- Engineered an automated pull request risk analysis pipeline that evaluates structural risk signals across open PRs and generates AI-assisted technical insights for engineering teams.
- Designed a deterministic multi-factor scoring engine analyzing 6 structural dimensions including complexity, review risk, volatility, and integration conflicts to compute composite risk scores (0–100).
- Integrated GitHub APIs to enrich PR metadata (files changed, commit density, reviewers, merge conflicts) enabling automated risk classification for repositories with multiple concurrent pull requests.
- Developed automated reporting workflows delivering color-coded risk summaries and executive engineering insights via Slack and HTML email reports.

SafeFlow: AI-Powered Crowd Safety Dashboard | *React, TypeScript, Vite, TailwindCSS* HackOdisha 5.0

- Developed a real-time crowd monitoring dashboard designed to integrate with AI-powered video analytics for early detection of stampede risks in large public gatherings.
- Built a modular React + TypeScript frontend supporting live video feeds, crowd analysis panels, and configurable data sources for real-time event monitoring.
- Implemented scalable UI components and reactive state management using React Hooks, enabling seamless integration with external AI prediction services.
- Designed the interface to support real-time safety alerts and predictive crowd-risk visualization for emergency response teams.

VortiqX: AI-Powered Security Vulnerability Scanner | *Python, Flask, OpenAI API, OWASP ZAP* 2025

- Built a web-based vulnerability scanning platform capable of analyzing websites for security misconfigurations, SSL issues, and common web vulnerabilities.
- Implemented automated security checks including HTTP security headers validation, directory traversal testing, and XSS protection verification.
- Integrated AI-assisted vulnerability analysis to generate contextual security explanations and remediation recommendations.
- Designed a responsive interface enabling real-time URL security scans with categorized risk reporting.

AgroWatch: Farm Biosecurity Monitoring Platform | *React, TypeScript, Supabase, TailwindCSS* SIH 2025

- Developed a digital monitoring platform for poultry and pig farms to support biosecurity compliance and disease risk tracking.
- Built a full-stack dashboard using React and Supabase enabling farm profile management, shed monitoring, and operational data tracking.
- Implemented secure authentication and real-time farm data management to support early detection of biosecurity risks in livestock environments.

ARBITICKER: Agentic AI Stock Decision Engine | *Python, FastAPI, JavaScript, Node.js, Vite, Groq API, DeepSeek AP*

- Built a multi-agent financial analysis system simulating a courtroom-style debate between Bull, Bear, and Quant agents before issuing investment verdicts.

- Developed a FastAPI backend exposing REST endpoints for stock analysis, profile creation, and watchlist management.
- Integrated Groq and DeepSeek LLM APIs to generate opposing financial perspectives before a deterministic rule-based judgment.
- Implemented a rule-based decision engine producing BUY/HOLD/SELL verdicts using quantitative thresholds derived from market data.
- Built a Vite-based frontend dashboard allowing users to analyze stock tickers and manage AI-assisted investment watchlists.
- Integrated the yfinance API to fetch real-time stock market data for quantitative evaluation.

EDUCATION

Amrita Vishwa Vidyapeetham	Kerala
<i>B.Tech in Computer Science and Engineering CGPA: 6.39</i>	<i>Aug 2024 – May 2028</i>
Sri Chaitanya IIT Academy	Madhapur, Hyderabad
<i>Higher Secondary Education</i>	<i>May 2022 – May 2024</i>
DAV PUBLIC SCHOOL	Hyderabad
<i>Primary and Secondary Education</i>	<i>May 2014 – May 2022</i>

RELEVANT COURSEWORK

Data Structures and Algorithms, Operating Systems, Internet of Things (IoT), Principles of Functional Programming (Haskell), Database Management Systems, Object-Oriented Programming